Disclaimer/credit. These questions have largely been ~~stolen~~ borrowed from natural sciences maths and vector calculus courses.

1)

Let $\boldsymbol{a}$ and $\boldsymbol{b}$ be fixed vectors. Show that $\boldsymbol{\nabla}(\boldsymbol{a}\cdot\boldsymbol{x}) = \boldsymbol{a}$. Evaluate the divergence and the curl of each of the following vector functions of $\boldsymbol{x}$:

$$\boldsymbol{x}\,, \qquad \boldsymbol{a}(\boldsymbol{x}\cdot\boldsymbol{b})\,, \qquad \boldsymbol{a}\times\boldsymbol{x}\,, \qquad \boldsymbol{x}/|\boldsymbol{x}|^3\,.$$

2. (i) Let $\phi(\mathbf{x})$ be a scalar field and $\mathbf{v}(\mathbf{x})$ a vector field. Show, using suffix notation, that

$$\nabla\,.\,(\phi\mathbf{v}) = \nabla\phi\,.\,\mathbf{v} + \phi\nabla\,.\,\mathbf{v}, \qquad \nabla\times(\phi\mathbf{v}) = \nabla\phi\times\mathbf{v} + \phi\nabla\times\mathbf{v}.$$

(ii) Evaluate the divergence and curl of each of the following:

$$(\mathbf{a}\,.\,\mathbf{x})\mathbf{b}, \quad \mathbf{a}\times\mathbf{x}, \quad r\mathbf{x}, \quad \frac{\mathbf{x}-\mathbf{a}}{|\mathbf{x}-\mathbf{a}|^3},$$

where $r = |\mathbf{x}|$ and $\mathbf{a}$, $\mathbf{b}$ are constant vectors.

(iii) The vector fields $\mathbf{F}$ and $\mathbf{G}$ are everywhere parallel, non-zero and solenoidal. Show that $\mathbf{F}\,.\,\nabla(F/G) = 0$, where $F = |\mathbf{F}|$ and $G = |\mathbf{G}|$.

3)

Use suffix notation to show that for vector fields $\mathbf{u}(\mathbf{x})$ and $\mathbf{v}(\mathbf{x})$,

$$\nabla\times(\mathbf{u}\times\mathbf{v}) = (\nabla\,.\,\mathbf{v})\mathbf{u} - (\nabla\,.\,\mathbf{u})\mathbf{v} + (\mathbf{v}\,.\,\nabla)\mathbf{u} - (\mathbf{u}\,.\,\nabla)\mathbf{v}.$$

Show also that $(\mathbf{u}\,.\,\nabla)\mathbf{u} = \nabla(\frac{1}{2}|\mathbf{u}|^2) - \mathbf{u}\times(\nabla\times\mathbf{u})$.

4) Trickier, note that solenoidal means $\nabla\cdot\mathbf{F} = 0$,

Suppose that $\mathbf{F}\colon \mathbb{R}^3 \to \mathbb{R}^3$ is a solenoidal vector field. Show that $\mathbf{F} = \nabla\times\mathbf{A}$ where

$$\mathbf{A}(\mathbf{x}) = \int_0^1 \mathbf{F}(t\mathbf{x})\times(t\mathbf{x})\,\mathrm{d}t.$$

5) For $r = |\mathbf{x}|$, verify, in ubdex notation, that

$$\nabla r^n = nr^{n-2}\mathbf{x}$$

6)

For electric and magnetic fields $\mathbf{E}(\mathbf{x},t)$ and $\mathbf{B}(\mathbf{x},t)$, define the quantities $U = \frac{1}{2}(\epsilon_0|\mathbf{E}|^2 + \mu_0^{-1}|\mathbf{B}|^2)$ and $\mathbf{P} = \mu_0^{-1}\mathbf{E}\times\mathbf{B}$. Use Maxwell's equations with $\mathbf{J} = \mathbf{0}$ to establish the conservation law $\partial U/\partial t + \nabla\,.\,\mathbf{P} = 0$.

7)

The velocity field $\mathbf{u}(\mathbf{x}, t)$ of an inviscid compressible gas obeys

$$\frac{\partial \rho}{\partial t} + \nabla . (\rho \mathbf{u}) = 0 \quad \text{and} \quad \rho\Big(\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} . \nabla)\mathbf{u}\Big) = -\nabla p,$$

where $\rho(\mathbf{x}, t)$ is the density and $p(\mathbf{x}, t)$ is the pressure. Show that

$$\frac{\partial}{\partial t}(\tfrac{1}{2}\rho u_i u_i) + \frac{\partial}{\partial x_i}(\tfrac{1}{2}\rho u_j u_j u_i + p u_i) = p \nabla . \mathbf{u} \quad \text{and} \quad \frac{\partial}{\partial t}(\rho u_i) + \frac{\partial t_{ij}}{\partial x_j} = 0$$

for a suitable symmetric tensor t_{ij}, to be determined.

8)

Using the transformation law for a second-rank tensor T_{ij}, show that the quantities

$$\alpha = T_{ii}, \quad \beta = T_{ij}T_{ji}, \quad \gamma = T_{ij}T_{jk}T_{ki}$$

are scalars, i.e., remain the same in all Cartesian coordinate systems. If T_{ij} is diagonal in some coordinate system, express the quantities above in terms of its eigenvalues. Hence deduce that the eigenvalues are the roots of the cubic equation

$$\lambda^3 - \alpha\lambda^2 + \tfrac{1}{2}(\alpha^2 - \beta)\lambda - \tfrac{1}{6}(\alpha^3 - 3\alpha\beta + 2\gamma) = 0.$$

9)

If $\mathbf{u}(\mathbf{x})$ is a vector field, show that $\partial u_i/\partial x_j$ transforms as a second rank tensor field. If $\sigma_{ij}(\mathbf{x})$ is a tensor field, show that $\partial \sigma_{ij}/\partial x_j$ transforms as a vector field.

10) Note: Part (ii) may be clearer after some matrices lectures

(i) By first decomposing into symmetric and anti-symmetric parts, show that an arbitrary second rank tensor T_{ij} can be written in the form

$$T_{ij} = \alpha\delta_{ij} + \epsilon_{ijk}\omega_k + B_{ij}$$

where α is a *scalar*, $\boldsymbol{\omega}$ a *vector* and B_{ij} a symmetric traceless second rank *tensor*.

(ii) Let $\mathbf{u}(\mathbf{x})$ be a vector field and let $T_{ij} = \partial u_i/\partial x_j$. Show that $\alpha = \frac{1}{3}\nabla . \mathbf{u}$ and $\boldsymbol{\omega} = -\frac{1}{2}\nabla \times \mathbf{u}$. Find B_{ij} in the case $\mathbf{u}(\mathbf{x}) = (xy^2, yz^2, zx^2)$: verify that $(0, 0, 1)$ is a principal axis at the point $\mathbf{x} = (2, 3, 0)$ and find the others.

11) Note: Part (ii) is trickier

(i) A tensor of rank 3 satisfies $T_{ijk} = T_{jik}$ and $T_{ijk} = -T_{ikj}$. Show that $T_{ijk} = 0$.

(ii) A tensor of rank 4 satisfies $T_{jikl} = -T_{ijkl} = T_{ijlk}$ and $T_{ijij} = 0$. Show that $T_{ijkl} = \epsilon_{ijp}\epsilon_{klq}S_{pq}$ where $S_{pq} = -T_{rqrp}$. [*Hint: Try to generalise a technique pertaining to antisymmetric second rank tensors.*]

12)

The array T_{ijk} is such that $T_{ijk}S_{jk}$ is a vector for every symmetric second rank tensor S_{jk}. Show by counterexample that T_{ijk} need not be a tensor; but show also that $T_{ijk} + T_{ikj}$ must be.

13)

In linear elasticity, the symmetric second rank stress tensor σ_{ij} (representing the forces on an elastic medium) depends on the symmetric second rank strain tensor e_{kl} (representing the corresponding deformation) according to $\sigma_{ij} = c_{ijkl}e_{kl}$. Assuming that c_{ijkl} is symmetric in k and l, explain why it must be a fourth rank tensor. For an isotropic medium, use the most general possible form for c_{ijkl} (which you may quote) to show that

$$\sigma_{ij} = \lambda\delta_{ij}e_{kk} + 2\mu e_{ij},$$

where λ and μ are scalars. Invert this equation to express e_{ij} in terms of σ_{ij}, assuming that $\mu \neq 0$ and $3\lambda \neq -2\mu$. Explain why the principal axes of σ_{ij} and e_{ij} coincide.

14)

Let $(x_0, x_1, x_2, x_3) \equiv (t, x_1, x_2, x_3)$ be coordinates for Minkowski space $\mathbb{R}^{1+3}$. Show that the two Maxwell equations $\nabla \times \mathbf{E} = -\partial\mathbf{B}/\partial t$, $\nabla \cdot \mathbf{B} = 0$, are equivalent to

$$\frac{\partial F_{ab}}{\partial x_c} + \frac{\partial F_{bc}}{\partial x_a} + \frac{\partial F_{ca}}{\partial x_b} = 0 \quad \text{where} \quad (F_{ab}) = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & -B_3 & B_2 \\ -E_2 & B_3 & 0 & -B_1 \\ -E_3 & -B_2 & B_1 & 0 \end{pmatrix}.$$

15)

In the following, the indices i, j, k, ℓ take the values $1, 2, 3$ and the summation convention applies.

(a) Simplify the following expressions:

$$\delta_{ij}v_j\,, \quad \delta_{ij}\delta_{jk}\,, \quad \delta_{ij}\delta_{ji}\,, \quad \delta_{ij}v_iv_j\,, \quad \varepsilon_{ijk}\delta_{jk}\,, \quad \varepsilon_{ijk}v_jv_k\,, \quad \varepsilon_{ijk}\varepsilon_{ij\ell}\,, \quad \varepsilon_{ijk}\varepsilon_{ikj}\,.$$

(b) Given that $A_{ij} = \varepsilon_{ijk}\,a_k$ (for all i, j), show that $2a_k = \varepsilon_{kij}\,A_{ij}$ (for all k).

(c) Show that $\varepsilon_{ijk}\,S_{ij} = 0$ (for all k) if and only if $S_{ij} = S_{ji}$ (for all i, j).